

Cost Effective Perovskite Solar Cell with Inorganic Hole-Transporting Material

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Abstract—Perovskite solar cell (PSC) with inorganic hole-transporting material is cost effective and stable. This paper presents a numerical analysis of $\text{CH}_3\text{NH}_3\text{PbI}_3$ based PSC using SCAPS-1D software. Actual absorption spectra of different layers of the PSC structure are considered in the analysis. The performance of the reference PSC has been investigated by varying the absorber layer thickness. It has been found that the efficiency is maximized when the absorber thickness is reduced. The doping densities of different layers are also varied to increase the power conversion efficiency (PCE) of the PSC. The proposed structure achieves a PCE of 14.63% where PCE increases by 1.9% in comparison with the reference cell. The proposed structure shows a short-circuit current (J_{sc}) of 21.17 mA/cm^2 , an open circuit voltage (V_{oc}) of 1.07V, and a fill factor of 64.38% under global AM 1.5 condition. The reduced absorber thickness, as well as the improved efficiency, makes the proposed PSC structure cost effective and efficient.

Index Terms—Perovskite solar cell, numerical analysis, power conversion efficiency, hole-transporting material.

I. INTRODUCTION

Today solar cells (SCs) are treated as the most eminent source of renewable energy as the demand of energy is increasing day after day all over the world. Recently, hybrid organo-halide perovskite has become a promising applicant as an absorber material for the future researches on SCs. The perovskite solar cell (PSC) has been widely investigated because of having high PCE, cost effectiveness and simple fabrication technique [1]. The stoichiometry of a halide perovskite is AMX_3 where A can be organic or inorganic ion (e.g., CH_3NH_3^+), M is a metallic ion (e.g., Pb^{2+}) and X refers to a halide ion. The first PSC was based on methylammonium lead halide ($\text{CH}_3\text{NH}_3\text{PbX}_3$) which was introduced in 2009 with low power conversion efficiency (~3.8%) and poor stability [2]. After years of active research, the $\text{CH}_3\text{NH}_3\text{PbX}_3$ based PSC has achieved a high PCE of ~22% [3]. The $\text{CH}_3\text{NH}_3\text{PbX}_3$ has excellent optoelectronic properties such as high absorption coefficient (10^3 – 10^4 cm^{-1}), high charge carrier mobility (2 – $66 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$), long carrier diffusion lengths (100 – 900 nm), low defect density, and high charge carrier life time ($\sim 270 \text{ ns}$) [4]. In addition, it has a tunable bandgap (1.55 – 2.5 eV), and the processing of $\text{CH}_3\text{NH}_3\text{PbX}_3$ from solutions can be performed at low temperatures ($<150^\circ\text{C}$) [4].

In a typical PSC, a thin perovskite layer is guided by two different layers such as HTM and ETM. Organic compound, 2,2',7,7'-tetrakis-(N,N-di-p-methoxy-phenylamine)-9,9'-spirobifluorene (spiro-MeOTAD) is the commonly used hole transporting material (HTM) for effective hole extraction from the absorber layer. However, spiro-MeOTAD is costly and degradation occurs due to moisture and heat [5]. Recently, Wu et al. have introduced $\text{Cu}_2\text{Zn}_2\text{Sn}_2\text{S}_4$ (CZTS) as a cost effective

HTM in PSCs [6]. The CZTS is an inorganic quaternary compound with excellent optoelectronic behavior. It has a direct tunable bandgap (1.4 – 1.5 eV), high hole mobility (6 – $30 \text{ cm}^2\text{V}^{-1}\text{s}^{-1}$), and high photon absorption coefficient ($\sim 10^4 \text{ cm}^{-1}$) [6]. It has been found that the CZTS improves the interface as a good contact between the electrode and the absorber layer and hence minimizes the hole surface recombination [6]. Hence it is an excellent absorber and HTM for perovskite based solar cells. However, it is necessary to analyze the PSC with CZTS as an HTM.

In this paper, a reference PSC structure (CZTS/ $\text{CH}_3\text{NH}_3\text{PbI}_3$ /TiO₂) has been simulated and compared with the published experimental results. A modified PSC structure is proposed to optimize the thickness of the $\text{CH}_3\text{NH}_3\text{PbI}_3$ layer. The doping densities of the $\text{CH}_3\text{NH}_3\text{PbI}_3$, TiO₂, and CZTS layers are also varied to improve the efficiency of the PSC.

II. NUMERICAL SIMULATION METHOD

The simulation has been performed using Solar Cell Capacitance Simulator (SCAPS-1D). The software is designed to simulate the solar cell based on the basic semiconductor equations i.e., the Poisson equation and continuity equation. Using SCAPS one can analyze J-V characteristic curve, PCE, FF, J_{sc} , and V_{oc} of the solar cell.

Figure 1 shows the reference PSC structure used in the simulation [6]. The multilayer PSC architecture is n-i-p planar where the intrinsic $\text{CH}_3\text{NH}_3\text{PbI}_3$ layer is inserted between the p-type CZTS and the n-type TiO₂. The back contact is gold (Au), and the front contact is fluorine doped tin oxide (FTO) which is a transparent conducting oxide (TCO) [6]. The series and shunt resistances of the reference cell are $9.8 \Omega\text{cm}^2$ and $920.8 \Omega\text{cm}^2$, respectively [6]. In addition, two hypothetical interface layers (IL) with high defect densities are inserted between TiO₂/ $\text{CH}_3\text{NH}_3\text{PbI}_3$ and $\text{CH}_3\text{NH}_3\text{PbI}_3$ /CZTS interfaces to consider the charge carrier loss due to recombination at the interfaces.

Table I shows the material parameters of different layers used in the numerical simulation. The parameters are selected based on the experimental data and the literature values [7–11]. The conduction band density of states and valance band density of states are considered to be $2.2 \times 10^{18} \text{ cm}^{-3}$ and $1.8 \times 10^{19} \text{ cm}^{-3}$, respectively for each layer [7]. At the back contact, the surface recombination velocities for both electrons and holes are assumed to be $1 \times 10^5 \text{ cm/s}$ and $1 \times 10^7 \text{ cm/s}$, respectively [7]. On the front contact, the surface recombination velocities for both electrons and holes are assumed to be $1 \times 10^7 \text{ cm/s}$ and $1 \times 10^5 \text{ cm/s}$, respectively. The defect type is assumed to be neutral. The capture cross section is considered to be $1 \times 10^{-15} \text{ cm}^2$ for both electrons and holes. The energetic distribution is Gaussian type having a characteristic energy of 0.1 eV [7]. The work functions of Au

and TCO are considered to be 5.1eV and 4.41eV, respectively [7]. The defect density for the $\text{CH}_3\text{NH}_3\text{PbI}_3$ are assumed to be $1.4 \times 10^{15} \text{ cm}^{-3}$ which gives the carrier diffusion lengths of electron and hole of .74 μm and .61 μm .

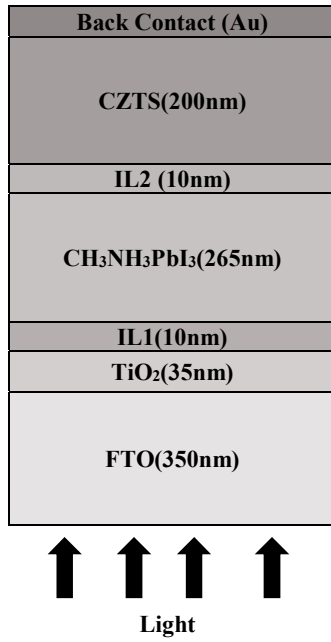


Fig. 1 The reference perovskite solar cell(PSC) structure [6].

Figure 2 shows the energy band diagram of the reference PSC structure. The energy band diagram is obtained from the output of the simulator. It is evident from Fig. 2 that both the FTO and TiO_2 have wider bandgap compared to $\text{CH}_3\text{NH}_3\text{PbI}_3$. Hence most of the incident photons are absorbed in $\text{CH}_3\text{NH}_3\text{PbI}_3$. As a result, J_{sc} and V_{oc} of the solar cell increase, that also increase the overall PCE of the solar cell.

Figure 3 shows the absorption coefficients of different layers used in the simulation. The absorption coefficients are taken from the literature [12–15]. From Fig. 3, it is evident that $\text{CH}_3\text{NH}_3\text{PbI}_3$ has high absorption coefficient at the visible spectrum. The absorption coefficient of CZTS is almost constant up to 1000 nm wavelength. However the absorption coefficient of CZTS falls sharply in the infrared region. The absorption coefficients of TiO_2 and FTO are negligible compared to $\text{CH}_3\text{NH}_3\text{PbI}_3$.

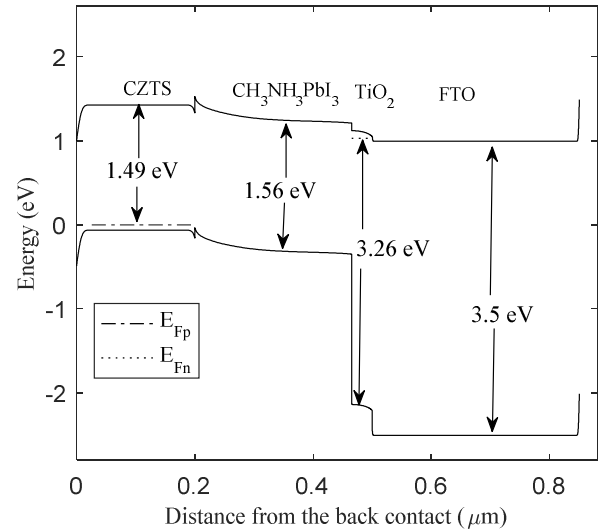


Fig. 2 Energy band diagram of the reference PSC.

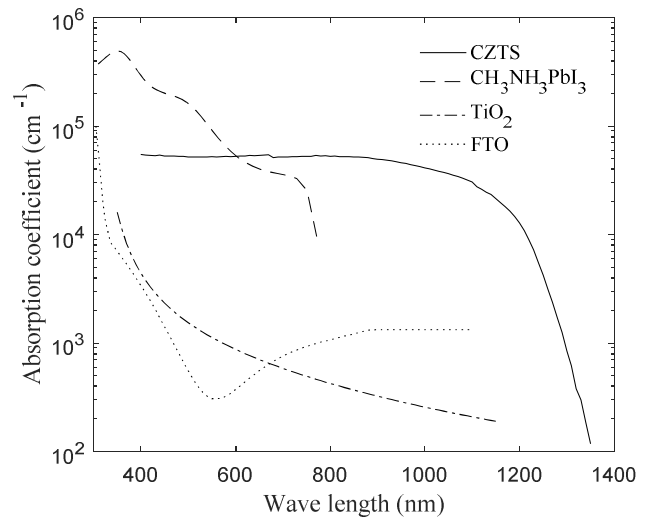


Fig. 3 Absorption coefficient of different materials of the reference PSC structure [12–15].

TABLE 1

THE MATERIAL PARAMETERS USED IN THE SIMULATION [7-11]

Parameters	FTO	TiO ₂	IL1	CH ₃ NH ₃ PbI ₃	IL2	CZTS
Thickness (nm)	350	35	10	265	10	200
Bandgap (eV)	3.5	3.26	1.56	1.56	1.56	1.49
Electron affinity (eV)	4	4	3.91	3.91	3.91	4.1
Relative permittivity	9	100	6.5	6.5	6.5	7
Electron mobility (cm ² V ⁻¹ s ⁻¹)	20	0.006	3	3	3	25
Hole mobility (cm ² V ⁻¹ s ⁻¹)	10	0.006	2	2	2	20
Donor density (cm ⁻³)	9×10^{17}	4.35×10^{17}	5.6×10^{15}	5.6×10^{15}	5.6×10^{15}	—
Acceptor density (cm ⁻³)	—	—	—	—	—	1.7×10^{18}
Defect density (cm ⁻³)	1.8×10^{18}	1×10^{14}	8.4×10^{20}	1.4×10^{15}	9.1×10^{21}	5×10^{15}

III. RESULT AND DISCUSSION

The simulation is performed under the global AM 1.5 condition and 300 K operating temperature. Figure 4 shows the J-V characteristic curve of the reference PSC structure. The solid line represents the simulation results and the circles represent the experimental data [6]. It is evident from Fig. 4 that the simulation results agree well with the experimental data. Table 2 shows the comparison between the experimental data and the simulation results.

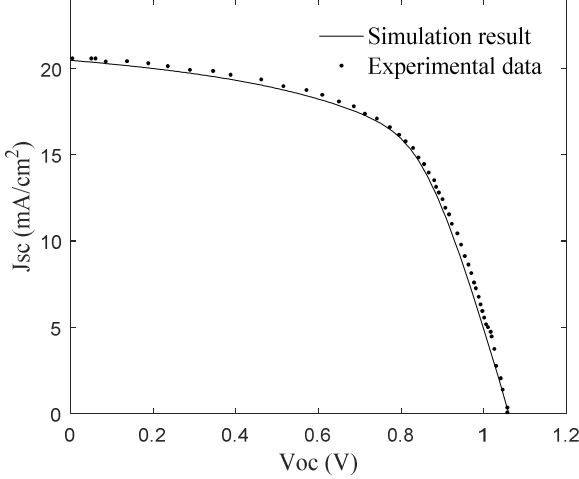


Fig. 4 J-V characteristic curve of the reference PSC. The solid line represents the simulation results and the circles represent the experimental data [6].

TABLE 2

COMPARISON BETWEEN THE SIMULATED RESULTS AND THE EXPERIMENTAL DATA [6]

Parameters	Simulation results	Experimental data
V_{oc} (V)	1.06	1.06
J_{sc} (mA/cm ²)	20.49	20.54
Fill Factor (%)	58.73	58.7
PCE (%)	12.76	12.75

The performance of a PSC depends on the thickness and the doping concentration of the absorber layer [7]. Figure 5 shows the device performance (V_{oc} , J_{sc} , FF, and PCE) of the PSC for various thickness of $\text{CH}_3\text{NH}_3\text{PbI}_3$ layer. The other parameters are the same as in Table 1. As shown in Fig. 5, V_{oc} increases when the absorber thickness increases. The V_{oc} reaches its maximum value at ~380 nm thickness, and then decreases slightly. As the absorber thickness increases, the charge carrier generation rate increases due to more light absorption. The V_{oc} increases when the thinner absorber establishes the generation rate higher than the recombination rate. The V_{oc} decreases when the reverse is true with thicker absorber [7].

Figure 5 shows that J_{sc} increases with the absorber thickness. The J_{sc} attains a maximum value at ~350 nm absorber thickness and then decreases. As the thickness increases more light absorbs and more excess carrier generates, which increases the J_{sc} . When the absorber thickness becomes greater than the diffusion length of the charge carrier, more recombination occurs which reduces the J_{sc} . From Fig. 5, it is evident that FF decreases continuously with the absorber thickness. Figure 5 also

shows that the PCE reaches its maximum value when the absorber thickness is ~235 nm. As the absorber thickness increases, more photon absorbs, and hence the PCE increases. When the absorber thickness is beyond the optimal value, carrier transport is limited by the charge carrier recombination due to more excess carriers and more traps. Therefore, the PCE decreases with the further increase in the absorber thickness. The result is consistent with the observation found in the literature [7].

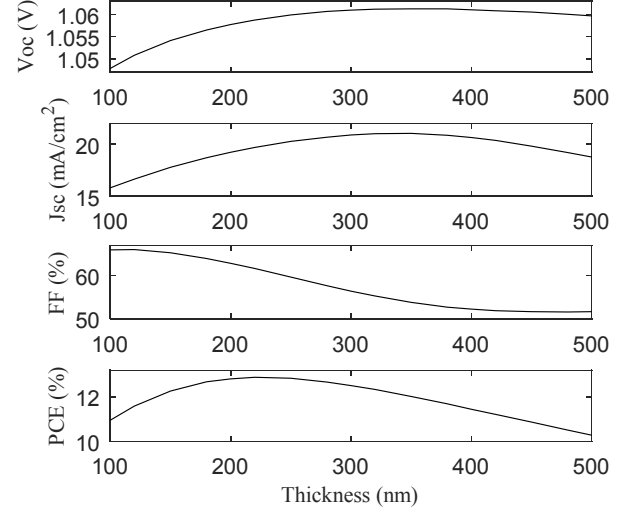


Fig. 5 Effects of the $\text{CH}_3\text{NH}_3\text{PbI}_3$ layer thickness on the performance of the PSC.

The cell performance is also affected by the doping densities of the HTM and ETM layers. Figure 6 shows the PCE of the PSC for various doping densities of the absorber, ETM, HTM layers. The other parameters are the same as in Table 1. It is evident from Fig. 6 that the efficiency of the PSC is almost constant as the doping of $\text{CH}_3\text{NH}_3\text{PbI}_3$ varies up to $4 \times 10^{14} \text{ cm}^{-3}$. The efficiency of the PSC decreases for heavily doped $\text{CH}_3\text{NH}_3\text{PbI}_3$. Figure 6 also shows that the efficiency increases as the doping densities of CZTS and TiO_2 increase. The doping increases the charge density, which enhances the conductivity of the layer and improves the device performance. However, the carrier mobility is limited by ionized impurity scattering due to heavy doping. Heavy doping also introduces immobile ions and hence generates deep traps [7]. Therefore, an appropriate doping level for different layers of the PSC has to be determined.

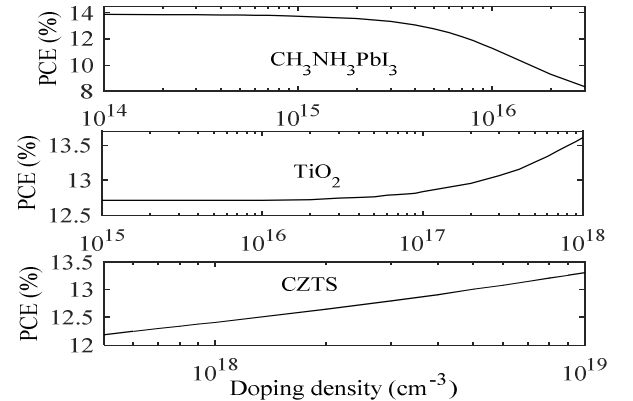


Fig. 6 PCE versus doping density of the absorber, ETM, and HTM layers.

Based on the analysis mentioned above a modified PSC structure has been proposed. Table 3 shows the proposed parameters of the modified PSC structure. The other parameters are the same as in Table 1. The thickness of the $\text{CH}_3\text{NH}_3\text{PbI}_3$ layer has been reduced. The doping densities of $\text{CH}_3\text{NH}_3\text{PbI}_3$, CZTS, and TiO_2 have been changed slightly. Figure 7 shows the comparison of the J-V characteristic of the proposed structure with the reference PSC. The solid line represents the J-V curve of the proposed structure and the dotted line represents the J-V curve of the reference PSC. The efficiency of the proposed PSC is 14.63% with $V_{oc} = 1.07\text{V}$, $J_{sc} = 21.17\text{ mA/cm}^2$, and $\text{FF} = 64.38\%$. The PCE of the proposed structure increases by 1.9% compared to the reference cell.

TABLE 3

THE PROPOSED DEVICE PARAMETERS

Parameters	Materials	Reference Structure	Proposed structure
Thickness (nm)	$\text{CH}_3\text{NH}_3\text{PbI}_3$	265	250
Doping density (cm^{-3})	$\text{CH}_3\text{NH}_3\text{PbI}_3$	5.6×10^{15}	3×10^{14}
	TiO_2	4.36×10^{17}	1×10^{18}
	CZTS	1.7×10^{18}	6.0×10^{18}

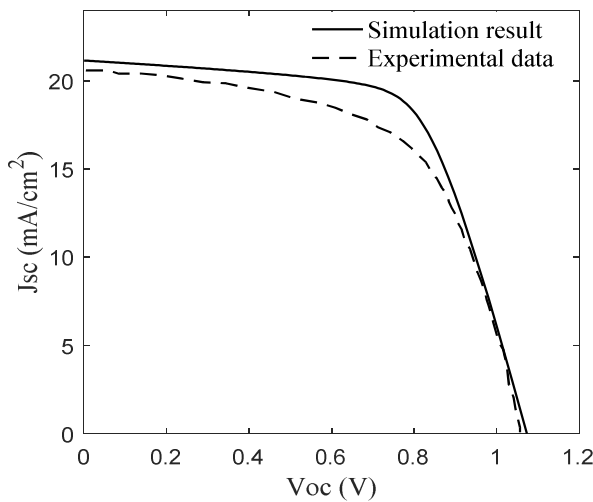


Fig. 7 Comparison of the J-V characteristic of the proposed structure with the reference PSC.

IV. CONCLUSIONS

A numerical model of a cost effective PSC structure has been developed using SCAPS-1D device simulator. The simulation results agree well with the published experimental data. The thickness of the absorber layer of the reference PSC has been reduced by 15 nm to get the maximum efficiency. The doping densities of $\text{CH}_3\text{NH}_3\text{PbI}_3$, CZTS, and TiO_2 are proposed to be $3 \times 10^{14}\text{cm}^{-3}$, $1 \times 10^{18}\text{cm}^{-3}$, and $6 \times 10^{18}\text{cm}^{-3}$, respectively to enhance the PCE of the reference PSC. The efficiency of the proposed PSC is 14.63% with $V_{oc} = 1.07\text{V}$, $J_{sc} = 21.17\text{ mA/cm}^2$, and $\text{FF} = 64.38\%$. The efficiency of the proposed PSC structure increases by 1.9% compared to the reference cell. The reduction of the absorber layer thickness makes the proposed PSC structure cost effective.

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